

Arab Republic of Egypt
Ministry of Education and Technical Education
Applied Schools Management Unit
الإمتحان التجريبي الثاني للصف الثالث
Advanced Physics-Year 2025-2026



NAME:

CLASS:3B

Group 1 : Choose the correct answer (20 min)

No	Statement
1	Which rule is used to determine the direction of force on a current-carrying conductor? a) Ampere's right hand rule b) Fleming's left hand rule c) Fleming's right hand rule d) Maxwell's cork-screw rule
2	When a conductor connected to an electrical circuit moves in a magnetic field, the direction induced current depends upon..... a) direction of the magnetic field and direction of motion of the conductor b) number of turns and length of the conductor c) strength of the magnetic field d) speed of movement of the conductor
3	In an ideal step-up transformer, which of the following statements is incorrect? a) The current ratio of secondary to primary is smaller than one. b) The turn's ratio of primary to secondary is smaller than one. c) The voltage ratio of primary to secondary is greater than one. d) The output-to-input electric power ratio is smaller than one.
4	The failure of classical physics in explaining black body radiation is mainly due to its inability to explain: a) Infrared radiation b) Ultraviolet radiation c) Decrease of intensity at high frequencies d) Continuous spectrum
5	In a metal, electrons are free to move but cannot escape because of: a) Their large mass b) Surface attractive forces c) Low temperature d) Magnetic forces
6	Electrons in the Coolidge tube are emitted due to: a) Photoelectric effect b) Thermionic emission c) Nuclear decay d) Ionization
7	Population inversion means: a) More atoms in ground state than in excited state

Administration and Operation Unit of Applied Technology Schools

	b) Equal atoms in ground and excited states c) More atoms in excited state than in ground state d) All atoms in excited states
8	At 0°K, a pure semiconductor behaves as: a) Good conductor b) Perfect insulator c) Semiconductor d) Superconductor
9	The movement of charge carriers across the junction due to concentration difference is called: a) Drift b) Radiation c) Conduction d) Diffusion
10	The basic idea of the transistor action states that a small change in base current causes: a) Large change in collector current b) Small change in collector current c) Small change in emitter current d) Large change in emitter current

Group 2: A) Mention the Scientific Term (10 min)

No	Statement	Answer
11	It is the minimum energy needed to free (release) the electrons from the surface of the metal.	
12	The radiation is almost one wavelength, as can be measured by the very narrow spectral width.	
13	The number of bonds broken per second will be equal to the number of bonds mended per second.	
14	The type of connection where N-type is connected to the positive terminal and the P-type is connected to the negative terminal of a battery.	
15	It is the ratio between the collector current and the base current.	
16	All electronic systems deal with natural quantities and convert them to continuous electrical signals.	

Administration and Operation Unit of Applied Technology Schools

B) Mention one factor only depend on the following (10 min)		
No	Statement	Answer
17	The induced electromotive force in the dynamo.	
18	The kinetic energy of the electron in Cathode Ray Tube.	
19	The characteristic radiation of X-rays.	
20	The kinetic energy of the emitted electrons due to photoelectric effect.	

Group 3: Correct the underlined in the following (20 min)		
No	Statement	Answer
21	The maximum value of magnetic flux cutting a certain area when the magnetic flux lines are <u>parallel</u> to that area.	
22	The magnetic field density generated from a <u>solenoid</u> carrying-current is inversely proportional to the distance to that point.	
23	The mutual induction coefficient measured in <u>Ampere/second</u> .	
24	A laminated iron core is used in an electric transformer to <u>increase</u> eddy currents.	
25	According to classical explanation of Planck's distribution, the intensity of radiation increases with the increasing the frequency and becomes <u>zero</u> at high frequencies.	
26	The photocurrent increases with the intensity of the incident light when the photon energy is <u>less</u> than the work function.	
27	When a photon of high frequency from X-rays collides with a free electron, the photon frequency <u>unchanged</u> and its direction changed.	
28	By increasing the temperature in pure Silicon, the covalent bonds are broken and some of electrons are freed and leaves behind a vacancy in the broken bond called a <u>gap</u> .	
29	In reverse bias of P-N junction, the depletion layer is <u>narrow</u> because the field due to the battery will aid the internal field.	
30	<u>OR</u> gate gives a low output (0) only if all its inputs are high.	

Group 4: Compare between each of the following (20 min)

No	Comparison		
31 32	Point of comparison	DC Current	AC Current
	Frequency		
33 34	Point of comparison	Grid	Anode
	Function in C.R.T		
35 36	Point of comparison	Laser beam	Ordinary light
	Coherence		
37 38	Point of comparison	P-type semiconductor	N-type semiconductor
	Majority Charge Carriers		
39 40	Point of comparison	NAND Gate	NOR Gate
	Truth table of two inputs		

Group 5: Solve the following problems (40 min)

No	Problems
41 42	In an AC generator, a rectangular coil of cross-sectional area of 0.07 m^2 and 100 turns rotates at a rate of 3000 revolutions per minute in uniform magnetic flux of density 0.1 T. Calculate the maximum electromotive force induced in the coil. $\left(\pi = \frac{22}{7}\right)$
43 44	A body at temperature 1500°K radiates out maximum energy at the wavelength $20000\text{\AA}$. If the sun radiates out maximum energy at $5000\text{\AA}$, calculate the temperature of the sun.
45 46	The wavelength of a photon is $5 \times 10^{-7} \text{ m}$. What is the momentum of the photon? ($h = 6.63 \times 10^{-34} \text{ J.s}$)
47 48	An X-ray tube operates at a voltage of 20 kV. What is the minimum wavelength of the emitted X-rays? (Given: $h = 6.63 \times 10^{-34} \text{ J.s}$ - $C = 3 \times 10^8 \text{ m/s}$ - $e = 1.6 \times 10^{-19} \text{ C}$)
49 50	The current gain (β) of a transistor in common emitter mode is 50. Calculate the emitter current if the base current $50 \mu\text{A}$.

Group 1 : Choose the correct answer

No	Answer	No	Answer
1	b) Fleming's left hand rule	6	b) Thermionic emission
2	a) direction of the magnetic field and direction of motion of the conductor	7	c) More atoms in excited state than in ground state
3	d) The output-to-input electric power ratio is smaller than one	8	b) Perfect insulator
4	c) Decrease of intensity at high frequencies	9	d) Diffusion
5	b) Surface attractive forces	10	a) Large change in collector current

Group 2: A) Mention the Scientific Term

No	Answer
11	Work function
12	Monochromaticity
13	Thermal Equilibrium or Dynamic Equilibrium
14	Reverse Connection or Reverse Bias
15	The Current Gain (β)
16	Analog Electronics

B) Mention one factor only depend on the following

No	Answer
17	$emf = BAN2\pi f \sin\theta$ • Magnetic flux density • Coil's Area • Frequency • Number of turns of the coil
18	The potential difference between cathode and anode
19	The kind of the target element
20	• The frequency (energy) of the incident photon • The work function of the metal

Group 3: Correct the underlined in the following

No	Answer
21	perpendicular
22	wire
23	Henry = V.s/A
24	decrease
25	infinite
26	more
27	decreased
28	hole
29	wide
30	NAND

Group 4: Compare between each of the following

No	Comparison		
31 32			
	Point of comparison	DC Current	AC Current
	Frequency	It has no frequency $f=0$	It has a frequency $f\neq 0$
33 34			
	Point of comparison	Grid	Anode
	Function in C.R.T	Control the number of electrons in the electron beam	It is used to accelerate the electron beam towards the screen
35 36	Point of comparison	Laser beam	Ordinary light
	Coherence	Photons emanate coherently both in time and place, since they come out together at the same time, phase and in the same direction.	Photons propagate randomly or incoherently. They emanate at different instants of time, and have inconsistent and varying phase.

37 38	Point of comparison	P-type semiconductor		N-type semiconductor		
	Majority Charge Carriers	Holes		Electrons		

39 40	Point of comparison	NAND Gate			NOR Gate																											
	Truth table of two inputs	<table><tr><td>A</td><td>B</td><td>Output</td></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	A	B	Output	0	0	0	0	1	0	1	0	0	1	1	1	<table><tr><td>A</td><td>B</td><td>Output</td></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	A	B	Output	0	0	0	0	1	1	1	0	1	1	1
A	B	Output																														
0	0	0																														
0	1	0																														
1	0	0																														
1	1	1																														
A	B	Output																														
0	0	0																														
0	1	1																														
1	0	1																														
1	1	1																														

Group 5: Solve the following problems

No	Answer
41 42	$emf = BAN2\pi f$ $emf = 0.1 \times 0.07 \times 100 \times 2 \times \frac{22}{7} \times \frac{3000}{60} = 220 V$
43 44	$\frac{\lambda_{m1}}{\lambda_{m2}} = \frac{T_2}{T_1}$ $\frac{20000}{5000} = \frac{T_2}{1500}$ $T_2 = 6000^\circ K$
45 46	$P_L = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}}{5 \times 10^{-7}} = 1.326 \times 10^{-27} kg.m/s$
47 48	$\lambda_{min} = \frac{hc}{eV} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{1.6 \times 10^{-19} \times 20 \times 10^3} = 6.2 \times 10^{-11} m$
49 50	$\beta = \frac{I_C}{I_B}$ $I_C = 50 \times 50 \times 10^{-6} = 2.5 \times 10^{-3} A$ $I_E = I_B + I_C = 50 \times 10^{-6} + 2.5 \times 10^{-3} = 2.55 \times 10^{-3} A$